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A Comparative Analysis of Single Nozzle and Multiple Nozzles Arrangements for Syngas Combustion in Heavy Duty Gas Turbine

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Heavy duty gas turbines are the core components in the integrated gasification combined cycle (IGCC) system. Different from the conventional fuel for gas turbine such as natural gas and light diesel, the combustible component acquired from the IGCC system is hydrogen-rich syngas fuel. It is important to modify the original gas turbine combustor or redesign a new combustor for syngas application since the fuel properties are featured with the wide range hydrogen and carbon monoxide mixture. First, one heavy duty gas turbine combustor which adopts natural gas and light diesel was selected as the original type. The redesign work mainly focused on the combustor head and nozzle arrangements. This paper investigated two feasible combustor arrangements for the syngas utilization including single nozzle and multiple nozzles. Numerical simulations are conducted to compare the flow field, temperature field, composition distributions, and overall performance of the two schemes. The obtained results show that the flow structure of the multiple nozzles scheme is better and the temperature distribution inside the combustor is more uniform, and the total pressure recovery is higher than the single nozzle scheme. Through the full scale test rig verification, the combustor redesign with multiple nozzles scheme is acceptable under middle and high pressure combustion test conditions. Besides, the numerical computations generally match with the experimental results.
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Keywords: gas turbine, combustion, syngas fuel, single nozzle, multiple nozzles

Introduction

As the concerns of fossil fuel emission and issue of environmental protection increase, the requirement for cleaner and more efficient energy utilization is of significant importance. The IGCC system application could provide a clean and efficient pathway for coal resources which is abundant in China. Besides, other resources such as biogas and industrial furnace gas could also be subsidiary energy provision for the whole power system.

Heavy duty gas turbine has been commercially successful energy and power technology for many decades. It is a good choice for gas utilization with the characteristics of low pollution, high efficiency, and high reliability. However, the syngas from the IGCC system and other industrial process is mainly hydrogen-rich gas with low calorific value. Therefore, the operation strategy of the existing gas turbine system and the combustor should be modified or redesigned in order to adopt syngas fuel. It is becoming a hot issue with continuous effort from the research and industrial field since the last decade [1,2], which includes the blended combustion of hydrogen and methane, syngas fuel with hydrogen and carbon monoxide. Most research focuses on the combustion performance with some literature focusing on the reaction kinetics and combustion dynamics [3,4]. Several original equipment manufacturers (OEMs) have designed and tested the advanced gas turbine combustor for both natural gas and unconventional fuel application [5–8]. Based on the accumulated experience, the combustor for syngas use is still under development and optimization [9,10] especially for heavy duty gas turbine.

With the rapid development of computer resources and numerical methods, computational fluid dynamics (CFDs) simulation has become an essential procedure in the combustor design/redesign process [11–13]. Previous numerical researches are mainly focused on the model combustor, small prototype or even concept combustor design with chemical kinetics investigation [14–18]. Besides, the emissions can also be modeled and predicted by numerical methods for industrial practice [19]. In order to develop combustor technology for syngas utilization, this paper presents a CFD investigation of syngas/air mixture reaction in a heavy duty gas turbine combustor which is originally operated with natural gas and light diesel as fuel.

In this paper, a full scale heavy duty combustor was modeled and analyzed by commercial CFD software FLUENT. Two combustor head arrangements with single nozzle and multiple nozzles were conceived and proposed for syngas combustion. In order to reduce the computation cost and improve efficiency, only the 1/5 or 1/4 (72 or 90 deg of the full annulus) of the combustion chamber is simulated for the two redesign schemes. The modeling includes realizable k–ε turbulence model and steady flamelet combustion model. Probability density function (PDF) is adopted to consider the turbulence and combustion interaction.

As known, the low calorific value of natural gas is about 36 MJ/Nm^3 while the syngas is only about 6–10 MJ/Nm^3. The huge difference in the fuel property indicates that it is difficult to evaluate the loss and gain during the redesign process. Not only the combustor should be redesigned, but also the operating point of turbine and compressor should be adjusted or even redesigned. For this case, the whole system performance was traded off to accommodate the fuel change. While for this research, the focus is mainly on the selection of different schemes. This investigation focuses on the combustor internal reacting flow field, temperature

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distribution, and the overall combustion performance. The obtained numerical results could provide strong support for design consideration and experimental preparation.

Combustor Geometry Description and Nozzle Arrangements

The prototype combustor of this research is selected from one typical type heavy duty gas turbine [20], which is applied for natural gas and light diesel oil. The chosen combustor refers to a reversal can combustor type. In the gas turbine unit, the can combustors are connected by the crossfire tubes and arranged uniformly in the circumferential direction around the rotor axis. The original can combustor geometry is shown in Fig. 1. Every combustor has one cylindrical combustor liner inserted in the liner bushing forming an annular guiding flow channel. The compressed air will flow through the diffuser, cooling holes/dilution holes/secondary holes/primary holes on the combustor liner, and combustor head inlet. The internal volume of the combustor can be separated into primary zone, secondary zone, and the dilution zone. The fuel and air are mixed and ignited in the primary reaction zone first with some air supplemented in the secondary zone. Then the combustion hot gas is diluted and temperature distribution altered. A complex three-dimensional shape pipe denoted as transition piece will guide the hot gas into turbine components. Usually, the traverse shape of the transition piece changes gradually from circular shape to sector shape. As shown in Fig. 2, the combustor liner and the combustor head with multiple nozzle arrangements are assembled as illustrations without the casing and bushing covering.

The shape and size of the combustor casing, the size of combustor liner and transition piece are kept as the original design for natural gas application. To accommodate the syngas utilization, the combustor head and nozzle are redesigned. The film cooling holes arrays are kept the same, while the location and size of the primary, secondary, and dilution holes are modified in order to adjust the combustion performance.

The total areas of the primary holes, secondary holes, and dilution holes on the combustor liner are the same for both the single and multiple nozzles schemes. However, the total number of the three kind holes for single nozzle arrangement is 4/5 of that in multiple arrangements. In the multiple nozzle arrangement, there are five swirl nozzles on the outer ring of the combustor. To keep the design symmetrical, the holes numbers are multipliers of five. Besides, the film cooling holes design of the single nozzle arrangement is the same as multiple nozzles arrangement.

Taking the multiple nozzles arrangement combustor for illustration, there are two rows of primary holes with ten primary holes for each row. The first row of primary holes is designed with the diameter of 17.9 mm, while the holes diameter of the second row 13.4 mm. After the primary holes rows, the secondary holes row for subsidiary combustion includes five holes, with the holes diameter of 11.2 mm. The last row of dilution includes five holes in this case with the holes diameter 39.8 mm. When considering the circumferential location, the secondary holes and the dilution holes keep the same which means on the same axial line. The primary holes show 18 deg stagger relative to the secondary holes as shown in Fig. 2. There are 16 rows of cooling holes designed to cross the whole combustor liner to provide enough cooling. The number of cooling holes in each row changes from 32 to 230. In addition, the maximum diameter of the cooling hole is 6.3 mm, while the minimum is only 1.4 mm.

Limited modifications can be conducted to redesign or retrofit the combustor head, mainly the nozzle arrangements. To be more specific, the air and fuel swirlers, fuel injection locations, and cover cones should be deliberately validated to burn the low calorific syngas while maintaining the combustor outlet temperature. In order to adopt syngas fuel, two combustor design schemes were proposed including the single nozzle arrangement and multiple nozzle arrangements. The most notable difference between the two schemes is the combustor head and nozzle structure.

The detailed structure of the single nozzle combustor head is shown in Fig. 3. The nozzle is designed with one syngas passage and one central oil passage. In the current simulation, the oil nozzle is not considered. The syngas is injected into the reaction zone through several inclined holes which helps generate swirling syngas flow. The air swirler is equipped with several inlet-holes and swirl vanes which forms rotating air flow passages. The air and syngas would mix and react downstream of the nozzle. Since the heat load of the combustor head is relatively high, the cooling air is guided into the flat cover plate and then through the cooling holes on the cover cones.

The multiple nozzles combustor consists of five syngas nozzle circumferentially around the central axis and one oil nozzle in the center. Every syngas nozzle is independent and composed of outer air swirler and inner syngas swirler. The central nozzle oil injection is only for the ignition and start-up process while the central nozzle air swirler is always in service. Therefore, for this investigation, the oil injection is neglected while the central air swirler is modeled in the simulation. The combustor head structure for multiple nozzles arrangement is displayed in Fig. 4. Impingement cooling is adopted for the cooling design of the combustor head.

Numerical Modeling and Validation of Combustor

Numerical Modeling Methods. Taking into account that the combustor chamber (without the transition piece) structure is axisymmetrical, only the 1/5 or 1/4 (72 or 90 deg of the full annulus)

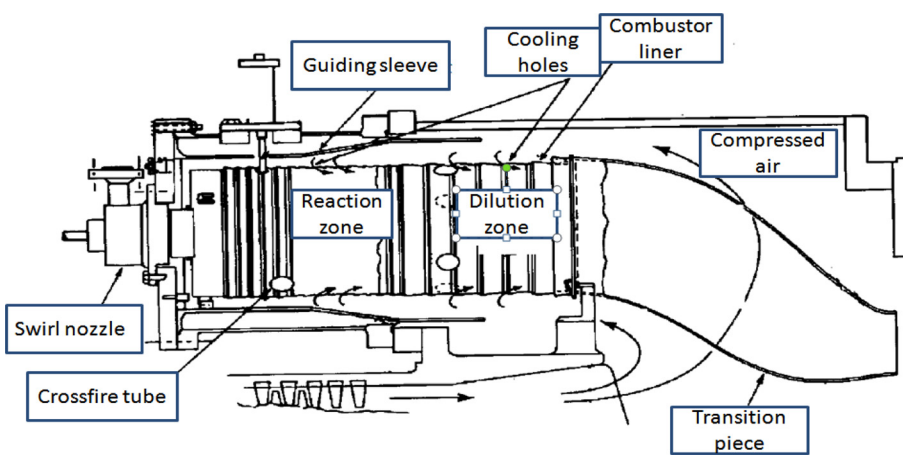

Fig. 1 Original combustor model structure

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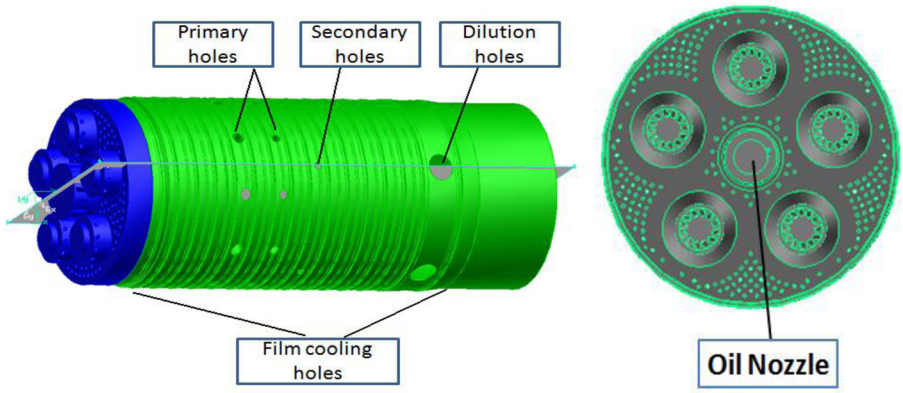

Fig. 2 Three-dimensional illustration of the combustor liner and the combustor head scheme

of the combustion chamber is modeled in order to improve the computational efficiency. The model revision and the mesh generation are conducted in the software GAMBIT. As introduced previously, there are 16 rows of cooling holes on the combustor liner, with the holes diameter ranging from 1.4 mm to 6.3 mm. In the simulation, the cooling holes are simplified as cooling slots, with the total area equal to that of cooling injection holes [21].The influence of the crossfire tubes on the combustor flow field is not considered in the computation. For the mesh independence check, two mesh schemes with 6×10^6 and 10×10^6 elements of mixed type were generated, respectively. In Fig. 5, the combustor liner and the cooling slots mesh on the periodic surface are given. Besides, the air and syngas swirler mesh is also displayed with tangential direction inlet.

For the reacting flow field simulation, the viscous N–S equations were solved and the results obtained. Since there exists strong swirling flow at the locations of the air swirler outlet and syngas fuel swirler outlet, the realizable k–ε turbulence model was selected [22]. For the wall region turbulence treatment, the standard wall function is chosen. The realizable k–ε model has been shown to provide more accurate prediction of high Reynolds number turbulent flows compared to the standard k–ε model which is also validated in Ref. [23]. Besides, the realizable k–ε model incorporates the factors of swirl and curvature which is more adaptable for the current simulation

$$\frac{\partial(\rho k)}{\partial t}+\frac{\partial(\rho k u_i)}{\partial x_i}=\frac{\partial}{\partial x_j}\left[\left(\mu+\frac{\mu_t}{\sigma_k}\right)\frac{\partial k}{\partial x_j}\right]+G_k+G_b-\rho\varepsilon-Y_M+S_k \tag{1}$$

$$\frac{\partial(\rho\varepsilon)}{\partial t}+\frac{\partial(\rho\varepsilon u_i)}{\partial x_i}=\frac{\partial}{\partial x_j}\left[\left(\mu+\frac{\mu_t}{\sigma_\varepsilon}\right)\frac{\partial\varepsilon}{\partial x_j}\right]+C_{1\varepsilon}+\frac{\varepsilon}{k}(G_k+C_{3\varepsilon}+G_b)-C_{2\varepsilon}\rho\frac{\varepsilon^2}{k}+S_\varepsilon \tag{2}$$

The combustion simulation was carried out using the steady flamelet combustion model. In steady flamelet combustion models, the variables such as temperature, density and species are specified by two parameters, the mixture fraction and scalar dissipation rate. The flamelet combustion model is suitable for diffusion combustion as introduced in the classic literature [24]. In the flamelet generation process, the opposing diffusion flame equations are transformed from physical space to mixture fraction space, with the scalar dissipation be modeled in the meantime

$$-\frac{\rho\chi_Z}{Le_i}\frac{d^2Y_i}{dZ^2}=\dot{\omega}_i \tag{3}$$

$$\rho\chi_Z\frac{d^2T}{dZ^2}+\frac{1}{C_p}\left(\frac{dC_p}{dZ}+\sum_{i=1}^{N}\frac{C_{pi}}{Le_i}\frac{dY_i}{dZ}\right)\frac{dT}{dZ}\rho\chi_Z=\frac{1}{C_p}\sum_{i=1}^{N}h_i\dot{\omega}_i \tag{4}$$

The detailed chemical reaction mechanism developed by Drake and Blint [25] is adopted which includes 22 species and is validated with the $CO/H_2/N_2$ fuel diffusion flame experimentin $CO/H_2/N_2$ laminar opposed-flow diffusion flames $CO/H_2/N_2$ laminar opposed-flow diffusion flames. Therefore, the mechanism is suitable for the simulation is considered in the current case. Through the calculation, the distribution of a variety of species can be obtained, such as H_2, CO, CO_2, NO, OH, etc. The probability density function (PDF) model is conducted to consider the interaction between turbulence and combustion. Radiative heat transfer was

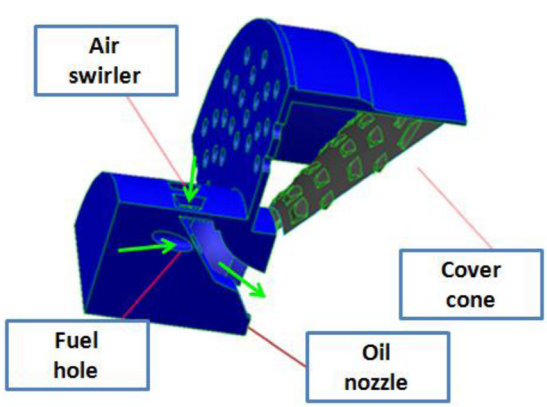

Fig. 3 Single nozzle combustor head structure

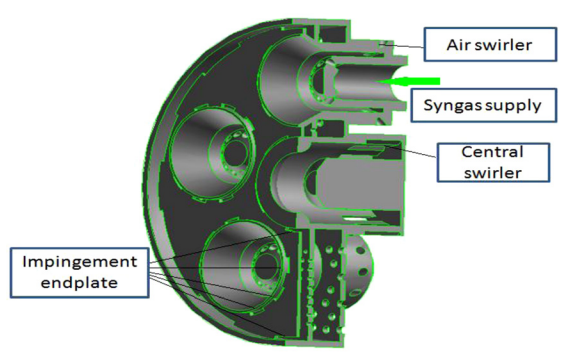

Fig. 4 Multiple nozzles combustor head structure

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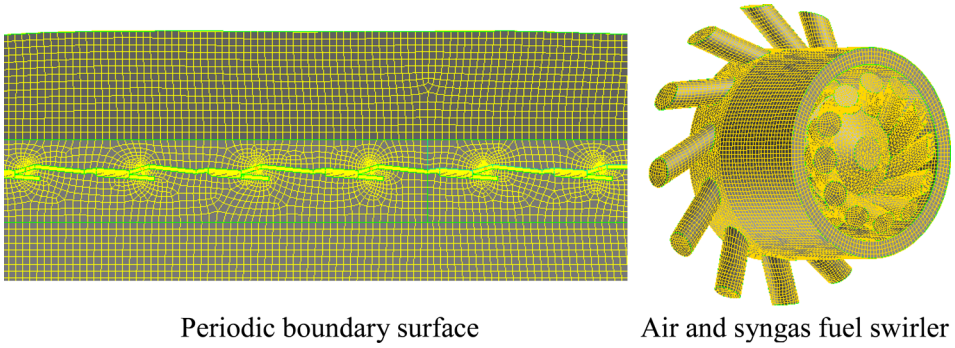

Fig. 5 Mesh generation for the combustor simulation

considered by using discrete ordinate radiation model. For the discretization and equation solving, the second-order upwind scheme and the SIMPLE method were selected to implement the calculation.

The external and internal sides of the combustor liner are directly connected by primary holes, secondary holes, dilution holes,, etc. Reversal flow supply from the compressor is set up from the downstream part. Boundary conditions include the air inlet mass flow rate, fuel inlet mass flow rate, periodic boundary, and combustor pressure outlet. The two sides of the sector as shown in Fig. 6 are periodic boundary. The outer wall is considered as adiabatic and the combustor liner wall is set as conjugate wall which allows for heat conduction.

The specified boundary conditions for the conversion of a single combustion chamber are as follows: air inlet mass flow rate is given as 13.63 kg/s and the inlet temperature is 355 °C. The pressure outlet is set up as 12.3 atm. For the syngas fuel, the inlet mass flow rate is given as 1.22 kg/s with the inlet temperature140 °C. The volume fraction of the syngas is shown in Table 1. The low calorific value of the syngas is 10 MJ/Nm^3.

Numerical Validation

Numerical method validation for the combustion simulation had been completed before the redesigned combustor was investigated. Reference [26] provides a full test data for a can combustor which adopts diffusion combustion. Therefore, the validation for the numerical method can be conducted first.

The combustor model for validation is displayed in Fig. 7 with the combustor length of 395 mm and the inner diameter of 125 mm. The combustor head is conical shape cover with air inlet holes on the wall [26].The inner diameter of the conical cover smaller end is about 25.4 mm, with the height of 28.6 mm. The flow paths of the air flow include six rows of air inlet holes on the conical cover, primary and dilution holes. Besides, several rows of film cooling holes are also considered. The fuel nozzle is located in the center with the diameter of about 7.67 mm. The fuel injection direction is vertical to the conical end. In the computation, 1/6 sector of the total cylinder is adopted to simulate the combustion. The domain covers the circumferential angle from −30 to 30 deg. The −30 and 30 deg planes denote the two periodic boundary planes, while the 0 deg plane denotes the central axial plane which cuts through the conical cover inlet holes, primary holes, and dilution holes. The axial flow direction is designating the z coordinate direction. All the numerical modeling methods have been mentioned above and the boundary types are also given in the figure.

Table 1 Detailed syngas composition

Composition	Volume fraction (%)
CO	47.8
H_2	37.3
CO_2	14.3
N_2	0.6

The validation simulation results were compared to the experimental test data in detail. As shown in Fig. 8, the axial velocity distributions at different axial locations were presented. The computation result is not ideally in good agreement with the experimental data however the general trend matches as shown in Fig. 8. The adopted technique in the literature is contact measurement which has disturbances on the flow field. Therefore, the measured velocity is also with some error especially in the place near the wall. Besides, the experimental test is divided into up and down parts. Although the measured two locations are symmetrical, the realistic measured data are not fully symmetrical while the computations are.

Based on the research group's work and accumulated knowledge, the numerical computations can predict the flow structure qualitatively right as shown in Fig. 9. Besides, the averaged outlet

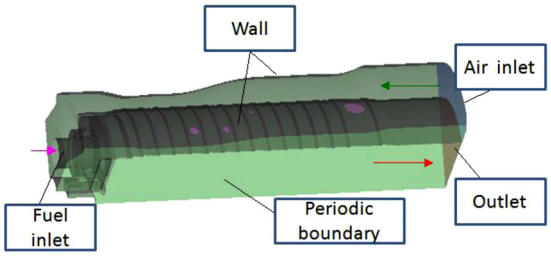

Fig. 6 Boundary conditions type specification

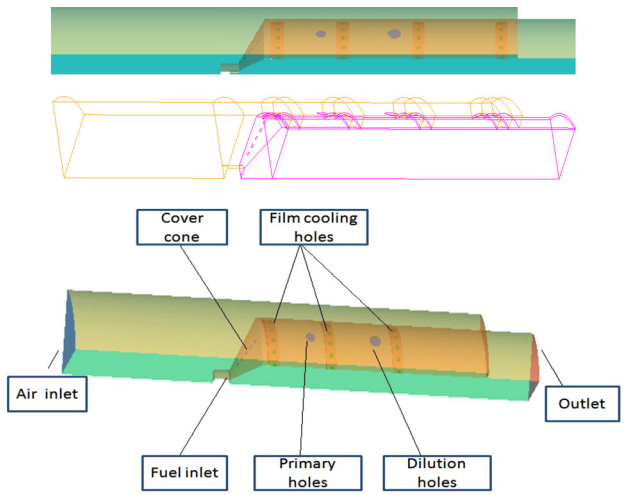

Fig. 7 Can combustor structure for combustion model validation

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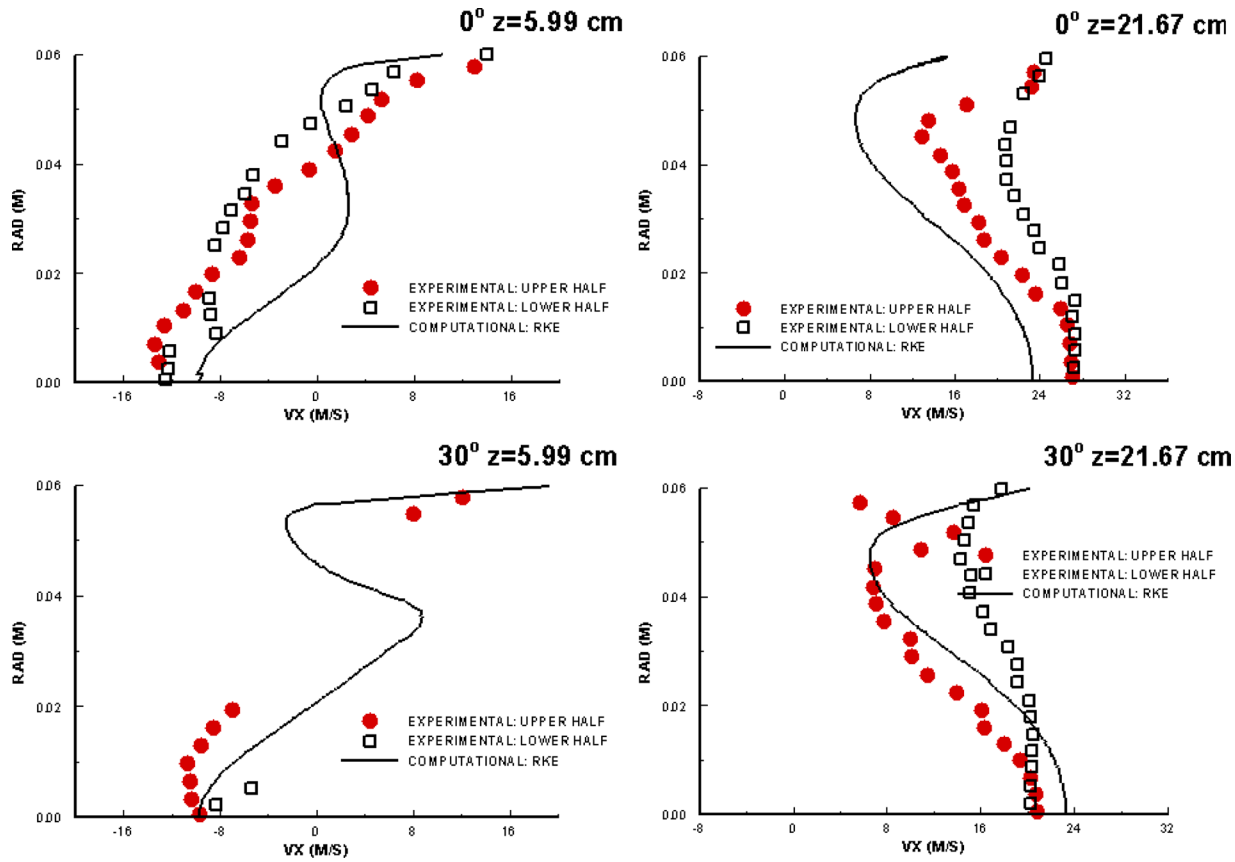

Fig. 8 Axial velocity comparison of numerical simulation and experimental results

temperature is 796.7 K, which is very close to the measured value 796 K. The computed total pressure loss is 3.68%, which is slightly larger than the measured data 3.28%. In general, the predicted overperformance of the combustor is taken with acceptable reliability.

Results and Discussion

The targeted combustor heat load corresponding to the specified boundary conditions is 13.33 MW. The total equivalence ratio is about 0.25 for the syngas fuel, with the combustor outlet temperature of about 1365 K. Since the combustor is designed to achieve nonpremixed combustion, the adiabatic flame temperature can reach over 2200 K at the flame front with equivalence ratio of about 1.

Before the computational results are discussed, the mesh independence check is conducted. The following figure gives a comparison of the computed results for the multiple nozzles case. The traverse cut plane at the dilution hole is chosen and the axial velocity contour is presented in Fig. 10. The mesh of 6×10^6 and 10×10^6 shows slight influences on the mixing result. Therefore, the mesh setup is destined with the total number 6×10^6.

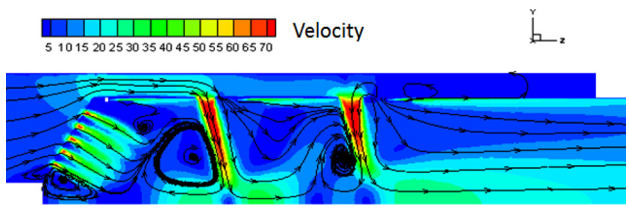

Fig. 9 Axial velocity distribution and streamline in 0 deg cutplane

Multiple Swirlers Arrangement

For the reversal flow combustor with a variety of holes on the liner, air flow distribution in the chamber is a significant issue for designer, which indicates excellent combustion and dilution performance. The mass flow rate distribution can be obtained by postprocessing the CFD simulation results, and the statistic data is listed in Table 2. According to the air splits statistic results, the single nozzle arrangement for syngas fuel is close to the natural gas original scheme. As the results indicated, the large variation of the air splits in single and multiple nozzles is mainly caused by the combustor head design and holes arrangements. However, the purpose of this investigation is to check the influences of different air splits on the flow field and combustion performance. Therefore, it is not considered to adjust the holes arrangement of multiple nozzles to match the single nozzle arrangement in this investigation. As shown in Table 2, the combustor head consumes about 14% of the total air mass flow for the multiple nozzles case while it is about 11% in the single nozzle case. Besides, the air flow for the primary/secondary/dilution holes and film cooling also show discrepancy between the two schemes. Although the total areas of the primary/secondary/dilution holes for the two schemes are the same, the number of the three type holes for multiple nozzles is 1.25 times of the single nozzle arrangement, which leads to smaller amount of air flow on the combustor liner. For the film cooling, the multiple nozzles arrangement consumes more air than the single nozzle design.

The total pressure recovery coefficient for the combustion chamber is shown in Table 3. Compared to the single nozzle combustor, the pressure recovery coefficient of the multiple nozzle arrangement is obviously improved, with about 2.1% enhancement. In contrast to the combustor utilizing natural gas and light diesel, the syngas fuel is of low calorific value. Therefore, the amount of syngas fuel admission is significantly larger than the traditional fuel type. When the fuel supply pressure is further

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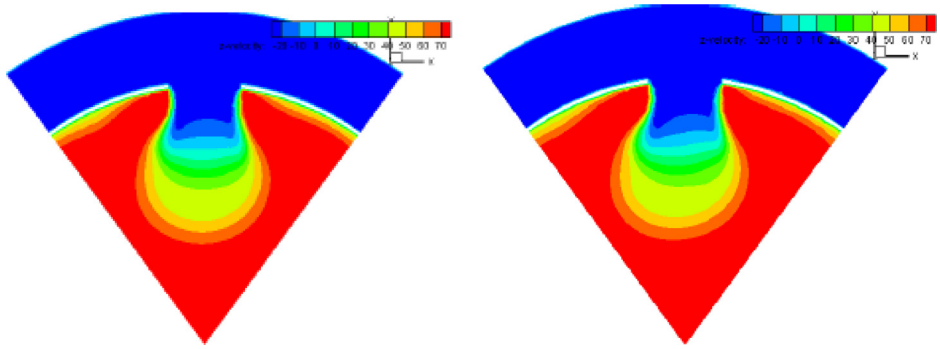

Fig. 10 Mesh independence check and comparison of axial velocity (left: 6×10^6, right: 10×10^6)

considered, the difference of pressure recovery coefficient for the two schemes will be even larger, reaching to about 3.1%.

The equations of the total pressure recovery coefficients C_{TPR1} and C_{TPR2} (with/without syngas fuel pressure modification) are listed as follows:

$$C_{\mathrm{TPR1}} = \mathrm{TP_{ou}}/\mathrm{TP_{in}} \tag{5}$$

$$C_{\mathrm{TPR2}} = \mathrm{TP_{ou}}/(\mathrm{TP_{in}} + V_{\mathrm{fuel}} * \mathrm{TP_{fuel}}/V_{\mathrm{air}}) \tag{6}$$

The general results show that the average temperature at the combustor outlet for the single nozzle and multiple nozzles arrangements are 1367 K and 1363 K, respectively, which is just slight difference. In order to give more detailed information of the two schemes, the temperature field, flow field structure, and composition distributions in terms of the computational results are given.

First, the axial cut-plane through the dilution holes is presented with the temperature distribution and the axial velocity distribution as shown in Fig. 11. As can be seen, for the single nozzle case, the high temperature zone is tending to be restricted in the area near the central axis of the combustion chamber. While the axial distance between the high temperature region and the combustor head is relatively large, the cooling requirement for the combustor head in the single nozzle case is not quite high. However, for the multiple nozzles case, the high temperature region possesses wider area before dilution. One thing is that the primary air is even closer to the combustor head which promotes the combustion reaction earlier in this case than the single nozzle case. Besides, the multiple nozzles mainly inject fuel on the outer side of the combustor head. Therefore, the high temperature zone expands radially and projects high heat load on the combustor head. In general, this is the reason why cooling design is more urgent in the multiple nozzles arrangement.

Furthermore, it can be inferred from Fig. 11 that there are more recirculation zones in the multiple nozzles case than the single nozzle case. In the single nozzle configuration, the recirculation zone is formed by the swirling air and syngas flow, and enhanced by the primary air injection. Basically, the zone is formed by the single swirler with some local corner vortex in the interface with the primary jets. For the multiple nozzles configuration, every single swirler unit can generate swirling recirculation zone. Besides, the central swirler can also creates the pilot swirling recirculation zone which helps enhance mixing and stabilize the flame. It is quite beneficial for the flame stability when there exist more recirculation zones in the flow field. From the formation mechanism of the recirculation, the single nozzle arrangement should adjust the cooperation of central swirler and the primary air injection. However for the multiple nozzles case, the single small swirler unit can be effective to generate recirculation. It is indicated that, for the multiple nozzles case, the design of the primary air jet location can be more flexible.

Besides, it is also displayed in the figure that the injection depth of secondary and dilution holes is larger in the single nozzle case than that of the multiple nozzles case. The reason is that the area of single dilution hole is larger in the single nozzle case. Deeper injection length is beneficial for the temperature distribution adjustment, and especially it can help reduce the peak value of outlet temperature for the single nozzle case. Further temperature uniformity could be realized by the transition piece guiding the flow after flowing through the combustor.

The mixture fraction distribution and the mole concentration of OH composition in the axial cut-plane are presented in Fig. 12 for both single nozzle and multiple nozzles arrangement. In the multiple nozzles case, the syngas fuel is injected through the single swirler unit around the axis which can realize good mixing in the short axial distance. The radial distribution of the mixture fraction is quite uniform without considering the central swirling air flow. For the single nozzle arrangement, the fuel mixing and transfer is mainly confined in the area near the combustor axis. Therefore, the mixing quality defines the flame front, combustion performance, and temperature distribution. From the distribution of OH composition mole concentration which can reflect the flame front location, the space for fuel heat release is more confined in the single nozzle case which also causes larger total pressure loss.

Full Scale Test Rig Introduction

In order to validate the combustor design, the full-scale high pressure/high temperature test rig is established and located in Lianyungang test base, which belongs to Energy and Power Research Center of Chinese Academy of Sciences as shown in Fig. 13. The whole test base includes all the infrastructures, subsidiary systems, and the test section. In detail, the experimental test base mainly includes air compressing system with heating system, high-pressure cooling water system, low pressure cooling water recycling system, water treatment system, steam boiler, fuel supply system (natural gas, synthetic gas, and light diesel oil), and

Table 2 Flow distribution over the combustor liner (percentages)

Type	Combustor	Big holes	Film cooling
Single nozzle	10.93	47.91	41.15
Multiple nozzles	14.42	39.38	46.20

Table 3 Total pressure recovery coefficient

Type	Without fuel pressure modification	With fuel pressure modification
Single nozzle	0.941	0.930
Multiple nozzles	0.961	0.959

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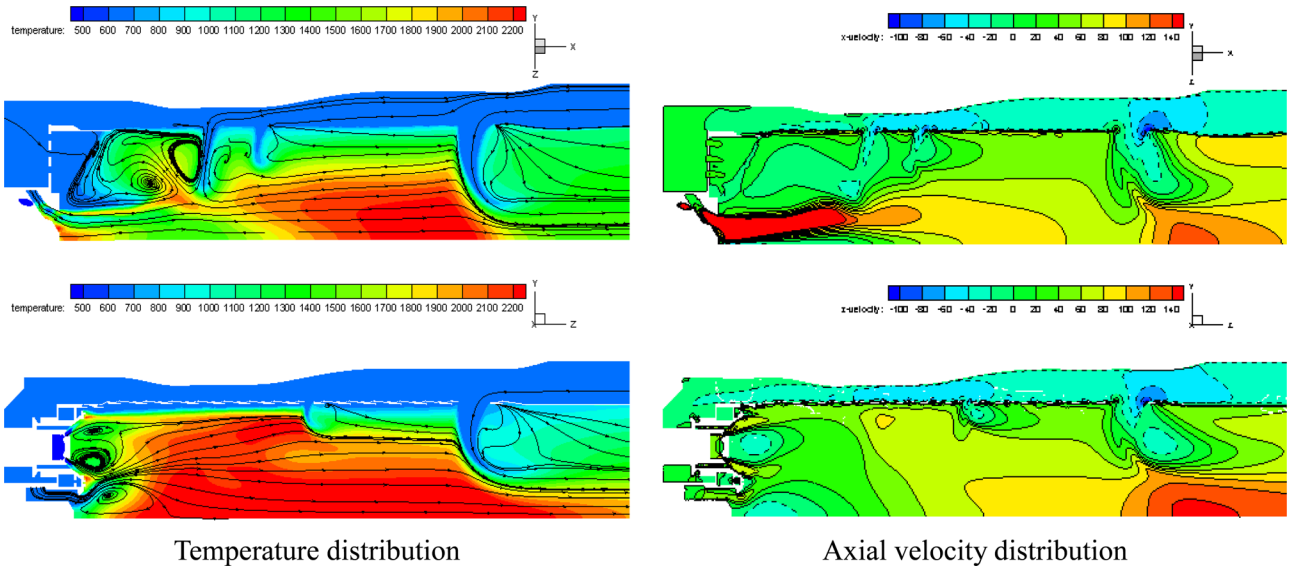

Fig. 11 Axial cut-plane of temperature and axial velocity distribution for two schemes (up: single nozzle case, down: multiple nozzles case)

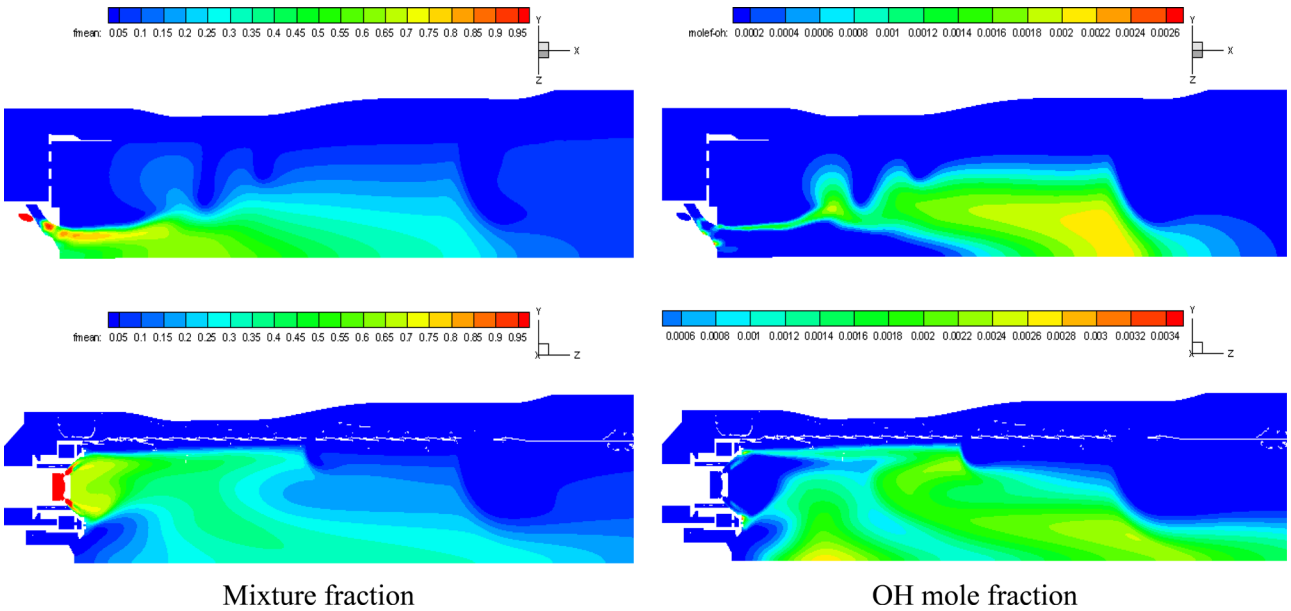

Fig. 12 Axial cut-plane of mixture fraction and OH mole fraction for two schemes (up: single nozzle case, down: multiple nozzles case)

Fig. 13 Display of full-scale combustion test facility

control measurement nitrogen and oxygen generation system. The experimental test base layout with the control station and data postprocessing methods can be referenced in the literature [11].The full-scale combustion test section is displayed in Fig. 11. The test base can provide test service for heavy duty gas turbine combustion test, in support of combustor design verification and validation. The capacity for compressed air supply can reach the maximum flow rate of 12 kg/s, with the maximum pressure 12 atm and the maximum temperature of 450 °C by electrical heating. The long term design target of the test rig capacity is over 20 kg/s with the maximum pressure 40 atm which is close to the most advanced technology level in the world.

Furthermore, the whole measuring system for the test section is illustrated. The inlet air temperature for the combustor is measured by five K-type thermocouples, with the average value of the measured data being adopted. The measurement error of the thermocouples is less than 1.0 °C with the sensitivity of about 0.1 °C. For the combustor outlet temperature at the transition piece

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Table 4 Flue gas analyzer precision

Composition	Measurement range	Measurement error
O_2	0–21 vol. %	±0.2%
NO	0–4000 ppm	±5 ppm (0–99 ppm), ±5% (100–4000 ppm)
NO_2	0–500 ppm	±5 ppm (0–99 ppm), ±5% (100–500 ppm)
CO	0–10,000 ppm	±10 ppm (0–199 ppm), ±5% (200–2000 ppm)

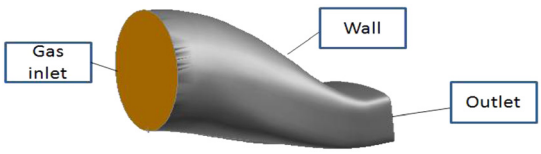

Fig. 14 Temperature distribution at the transition piece outlet

downstream, the measurement rake is installed with R-type thermocouple in the rakes. The rake can be driven by the stepping motor, scanning the whole temperature field at the transition piece outlet section. The measurement error of the R-type thermocouples is 1.0 °C with the sensitivity of about 1.0 °C. The combustor inlet air pressure and the combustion gas pressure at the transition piece outlet are measured by the calibrated total pressure probe, with the measuring accuracy of 0.1%. The inlet air mass flow rate is measured using the orifice plate flow meter supplied by the A+K Company. The fuel gas flow measurement is conducted by using vortex flow meter with the precision of 1%FS. The sampling probe for pollution emission is located at the exit of the transition piece, and the flue gas sampling analyzing device is Testo350 enhanced analyzer, with its precision details shown in Table 4.

Middle and High Pressure Test Verification

In this verification test, the multiple nozzles configuration is adopted and manufactured. Since the target pressure cannot be fully realized yet, the middle pressure combustion test is conducted first. Different from the design stage in which the fuel and air mixing and flow organization is significant, it is necessary to consider the transition piece in the verification stage. In this investigation, the geometry and the boundary condition type for the whole transition piece are displayed in Fig. 14. The test has also been completed under full-scale middle pressure combustion conditions. Test conditions are listed: inlet air pressure 2.3 atm, air preheating temperature 356 °C, air mass flow rate 2.2 kg/s, and syngas fuel mass flow rate 0.23 kg/s. The temperature of the syngas fuel is 140 °C. To save the computational cost, the gas inlet boundary conditions are transferred from the numerical results of combustor outlet. The acquired boundary variables include the temperature, velocity, and turbulent parameters distribution. Then the numerical iterations are completed only for transition piece

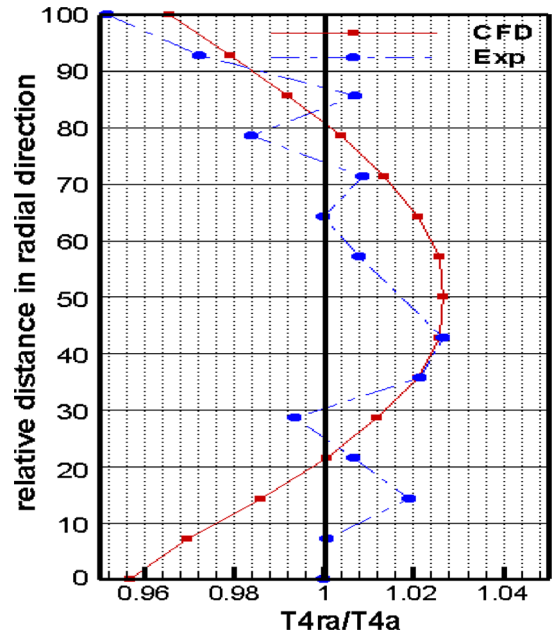

Fig. 16 Radial distribution of averaged temperature data

domain. As shown in Fig. 15, the temperature distribution of the numerical simulation and experimental results at the transition piece outlet are compared. Generally, the obtained numerical results are consistent with the measured value in trend. The temperature measurement for the transition piece outlet is conducted by installing the movable struts with thermocouples. All the measured surface data are not recorded at the same transient time step. In actual situation of the experiment, the temperature fluctuation is quite obvious which causes that the measured data is not as flat as the CFD results contour. The steady Reynolds-averaged Navier–Stokes simulations cannot predict the transient characteristic of the combustion condition. Besides, the temperature measurements strut can also influence the flow field which might cause the temperature distortion.

By comparing the normalization of the temperature distribution curve in Fig. 16, results show that numerical simulation can capture the maximum temperature spot and the general trend, which is helpful for the prediction and experimental work.

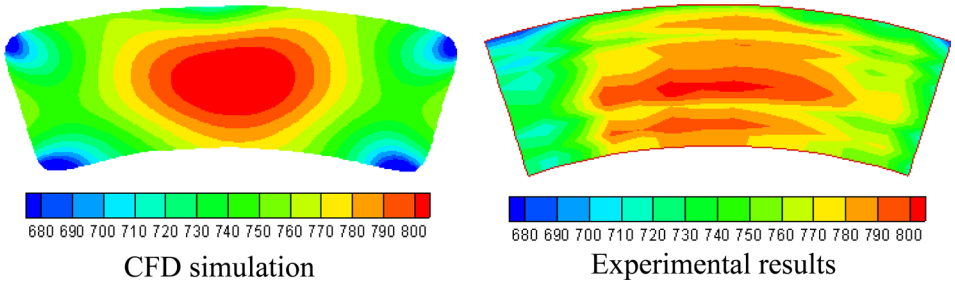

Fig. 15 Temperature distribution at the transition piece outlet

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The test has also been completed under full-scale high pressure combustion conditions. Detailed test conditions are as follows: inlet air pressure 10.02 atm, air preheating temperature 405 °C, air mass flow rate 10.54 kg/s, syngas fuel mass flow rate 1.062 kg/s, and the nitrogen mass flow rate 0.868 kg/s. The temperature of both the syngas fuel and nitrogen is 20 °C. The calorific value and compositions of the syngas fuel are shown in Table 1.Through the complicated combustion test process, the general combustor performance is acquired: the average temperature at the combustor exit of about 1216 °C, combustion efficiency reaching 99.9%, the total pressure recovery coefficient 0.955 without taking the fuel pressure into account, NO_x emissions 22 ppmv (at 16%O_2), CO emissions 17 ppmv (at 16%O_2).

After the combustion test, the detail check shows that there is no obvious high temperature region. The highest temperature spot of the combustor liner is below 861 °C without thermal barrier coating sprayed. Based on the industrial experience, if the thermal barrier coating is sprayed onto the liner, the highest temperature could be reduced further to 50 °C at least. Besides, the total pressure recovery coefficient of the numerical calculation is in good agreement with the experimental measurements. The full load combustion test is still under preparation.

Conclusion

Based on the combustion modeling validation, two design schemes of the full-scale heavy duty combustors with single nozzle and multiple nozzle arrangements were investigated. Numerical simulations using the commercial CFD software FLUENT were adopted. By comparing the numerical results of the two design arrangements, it is shown that the flow field structure in the multiple nozzles case is more preferable, with more uniform combustor temperature field and higher total pressure recovery coefficient.

In addition to the numerical simulation, the verification test is also completed for the multiple nozzles arrangement combustor. The test condition is full-scale middle (2.3 atm) and high pressure (10 atm). The combustor liner wall temperature, pollutant emissions, and total pressure recovery coefficient are all qualified for the combustor design proved by the test. In order to reach the design target and save the cost of time and resources, numerical simulations can be helpful to predict the flow field, temperature distribution, and overall performance in the combustor. Therefore, it is worth of conducting numerical simulations during the design and retrofit process.

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Nomenclature

C_{TPR1} = total pressure recovery coefficient without syngas fuel pressure modification
C_{TPR2} = total pressure recovery coefficient with syngas fuel pressure modification
TP_{fuel} = total pressure of syngas fuel
TP_{in} = total pressure of the inlet air
TP_{ou} = total pressure of the combusted gas at the combustor outlet
V_{air} = volume fraction of air in the combustor
V_{fuel} = volume fraction of syngas fuel in the combustor

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